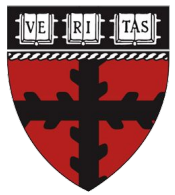
Lecture 22: Scaling: Kubernetes (cloud)

AC215/E115

Pavlos Protopapas
SEAS/Harvard



Outline

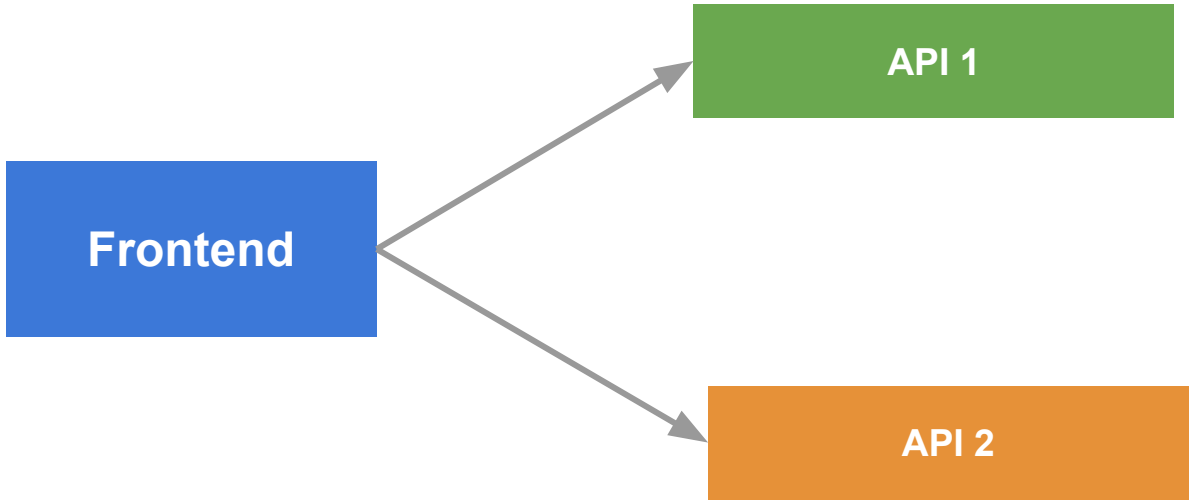
1. Motivation
2. Introduction to Kubernetes
3. Tutorial: Deploying a Kubernetes Cluster
4. Advantages of using Kubernetes

Outline

1. **Motivation**
2. Introduction to Kubernetes
3. Tutorial: Deploying a Kubernetes Cluster
4. Advantages of using Kubernetes

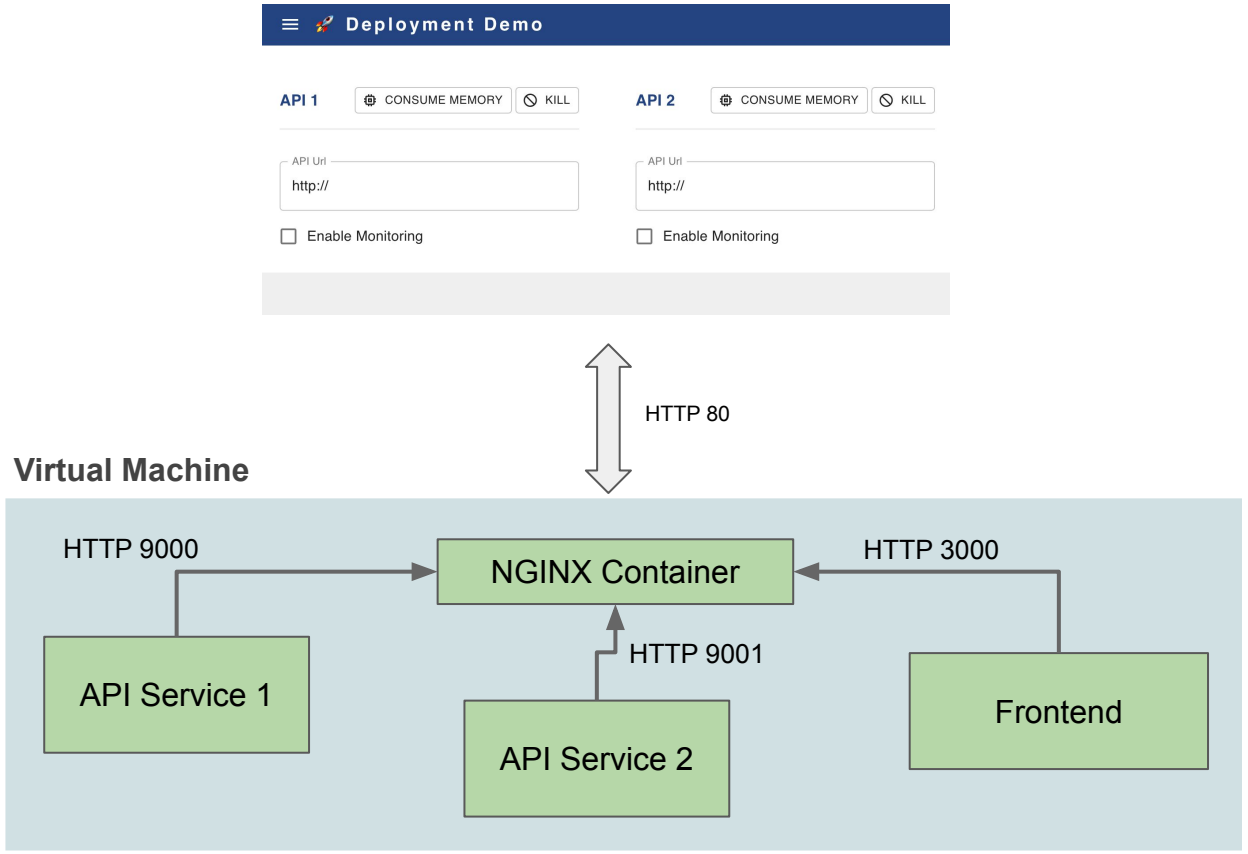
Motivation

Pavlos wants an app with 1 frontend & 2 backends



Motivation - 3 Containers in 1 VM

Support builds and deploys the app with the following architecture



Motivation - 3 Containers in 1 VM

Demo… [3 Containers in 1 VM]

Motivation - 3 Containers in 1 VM

Container Crashes

Pavlos must contact support for resolution.

Support Actions:

Access the server via SSH.

Perform the following fixes:

- Restart the container to reset memory.
- Relaunch a terminated container.

Motivation - 3 Containers in 3 VM

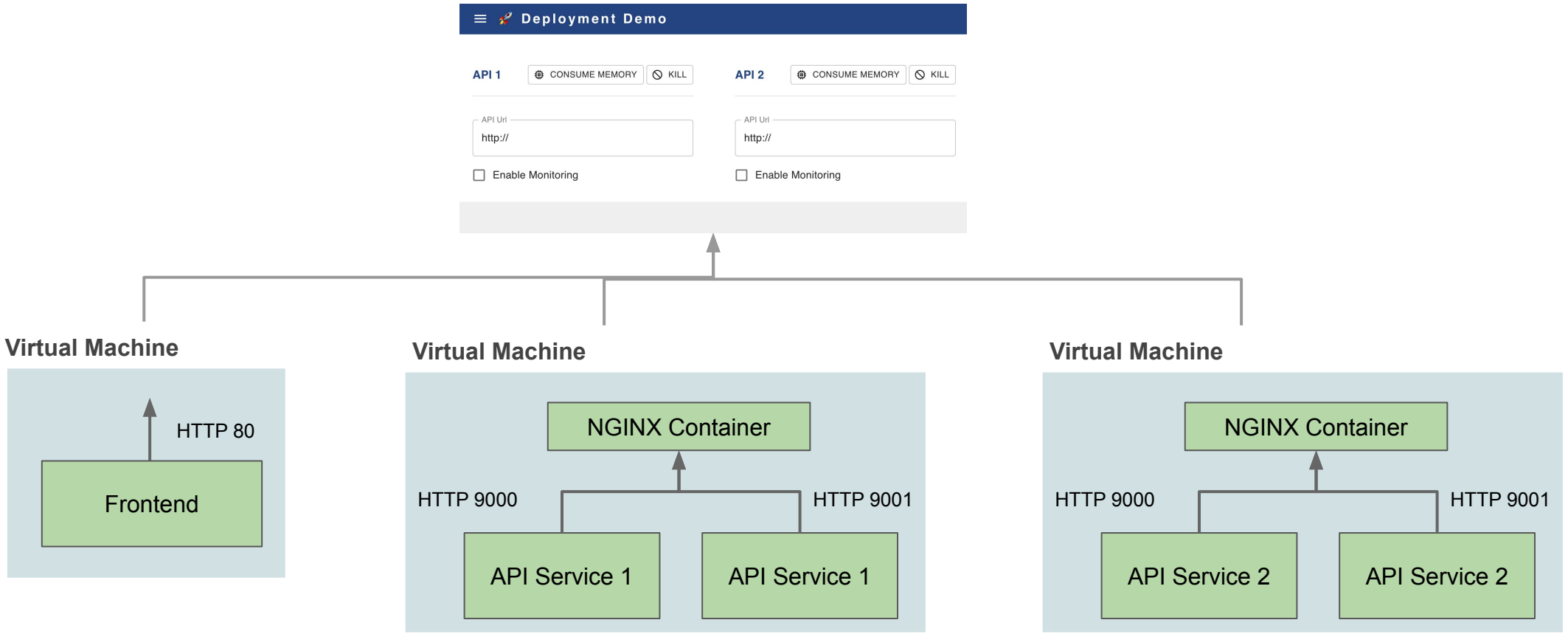
Pavlos' Request to *Support*

"Can we deploy the app across multiple servers?

This way, if one server goes down, I'll have a backup to rely on."

Motivation - 3 Containers in 3 VM

Support deploys the app on to 3 servers with backup apis



Motivation - 3 Containers in 3 VM

Demo… [3 Containers in 3 VMs]

Motivation - 3 Containers in 3 VM

Problems:

- When container crashes, Pavlos can switch to backup API manually
- *Support* SSHs into server and fix when available:
  - Memory reset with container restart
  - Startup a killed container

Motivation - Kubernetes

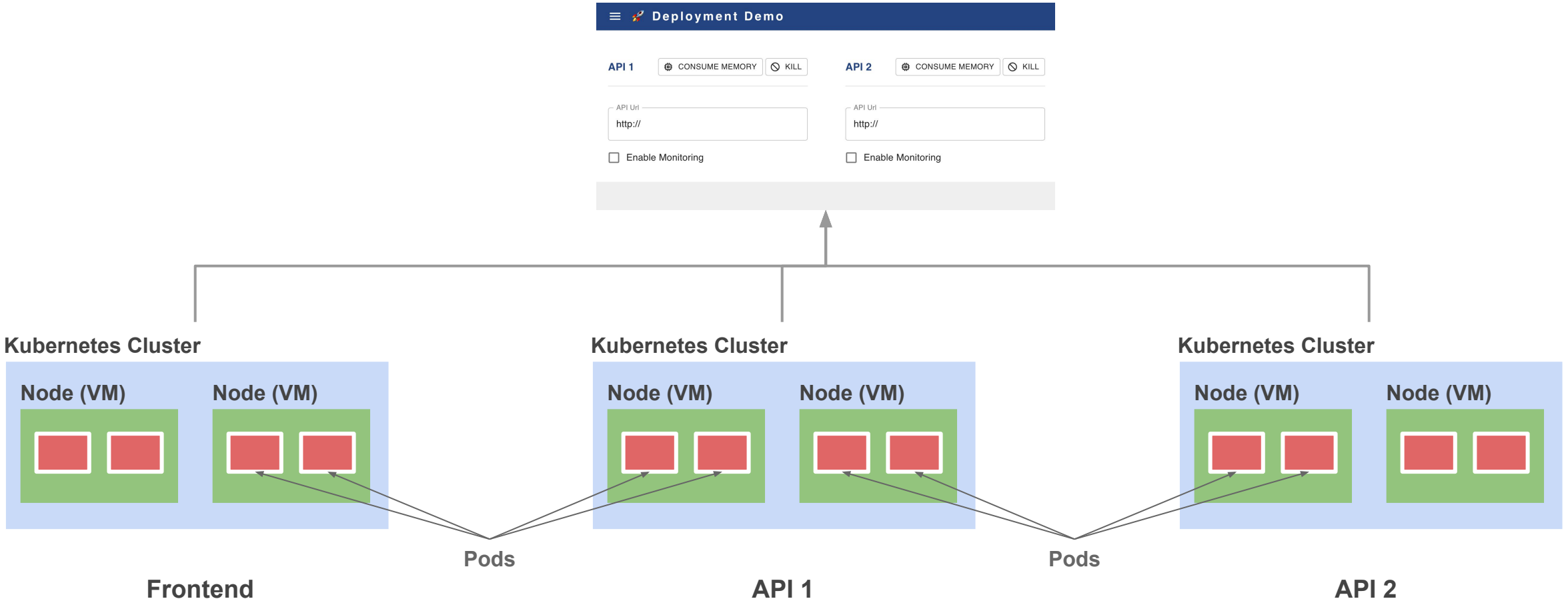
Pavlos' Question to Support

- Can we automate:
  - Failovers
  - Load balancing
  - Scaling
  - And other key processes?

Kubernetes to the rescue...

Kubernetes to the Rescue

Support deploys the app on to 3 k8s clusters with 2 nodes each



Kubernetes to the Rescue

Demo… [Kubernetes Cluster]

Kubernetes

Pavlos requests on automation:

- ✔ Failovers
- ✔ Load balancing
- ✔ Scaling

Outline

1. Motivation
2. **Introduction to Kubernetes**
3. Tutorial: Deploying a Kubernetes Cluster
4. Advantages of using Kubernetes

Kubernetes (K8s) to the Rescue



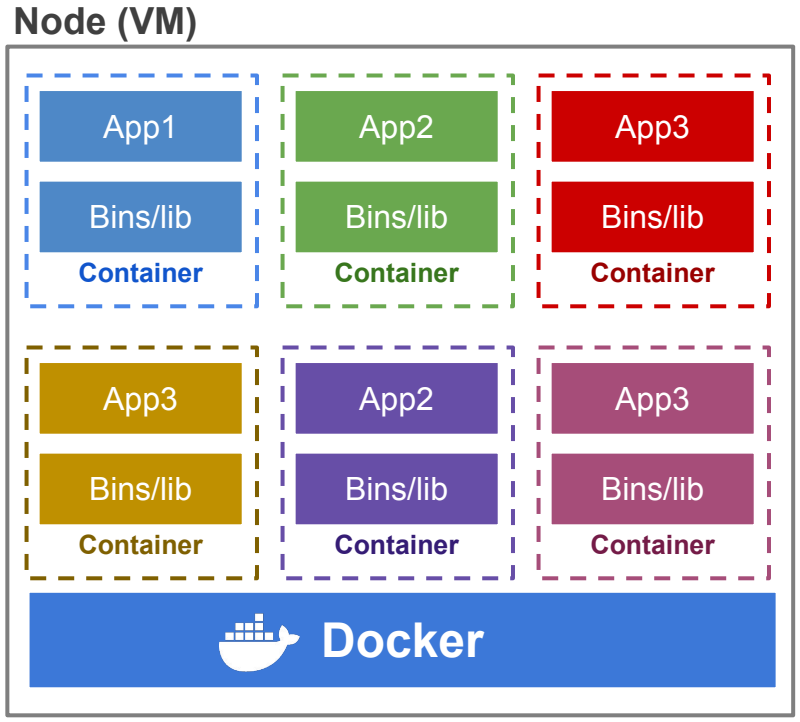
- K8s is an orchestration tool for **managing distributed containers** across a cluster of nodes (VMs).
- The word Kubernetes comes from the ancient Greek word **kubernḗtēs**, which means helmsman or pilot. The name is a reference to the role of a **helmsman**, who steers a ship and maintains a steady course.
- Kubernetes was announced by Google on June 6, 2014. The project was conceived and created by Google employees Joe Beda, Brendan Burns, and Craig McLuckie.

Kubernetes (K8s) to the Rescue

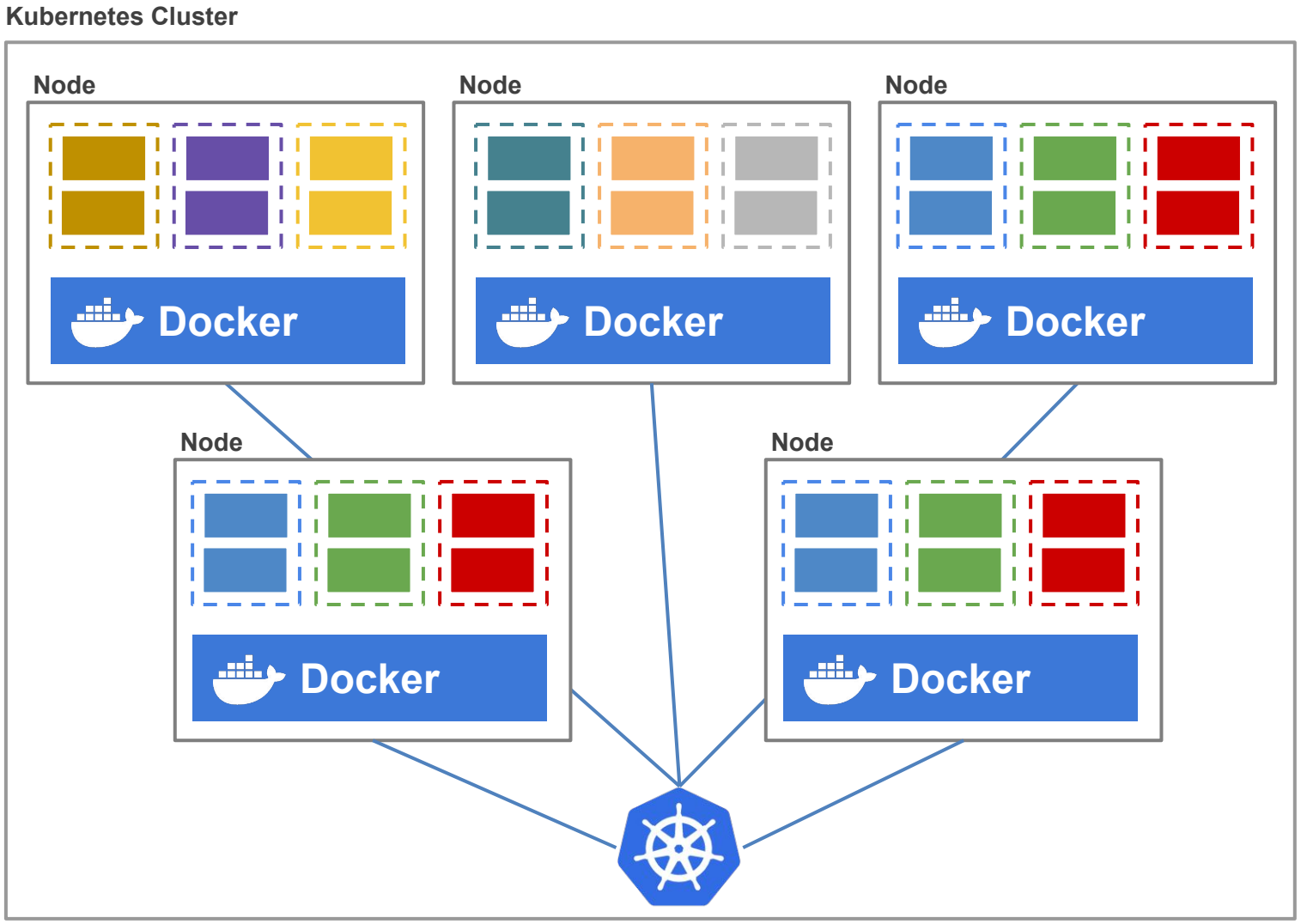


- Kubernetes (K8s) is made up of building blocks that help deploy and scale applications based on CPU, memory, or custom metrics.
- K8s itself follows a **primary-replica architecture** with components that govern an individual node and others part of the **control plane**
- Core concepts in Kubernetes include pods, services and deployments.
- K8s **users define rules** for how container management should occur, and then K8s handles the rest!

Container vs Kubernetes Deployment



Container Deployment



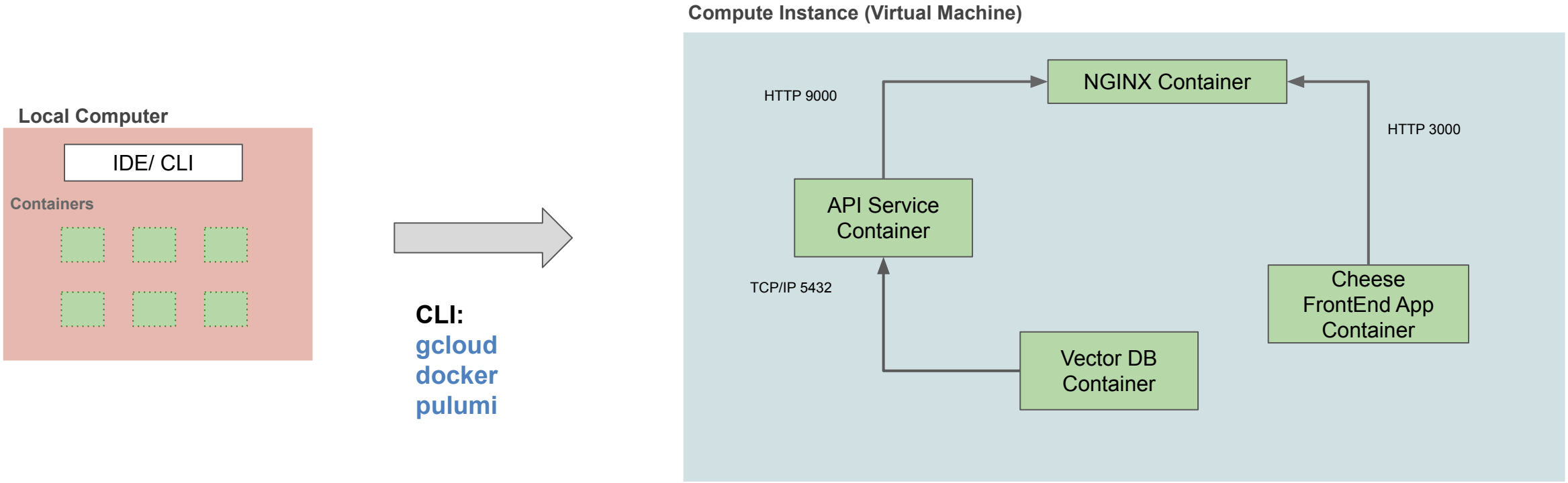
Kubernetes Deployment

Why Kubernetes?

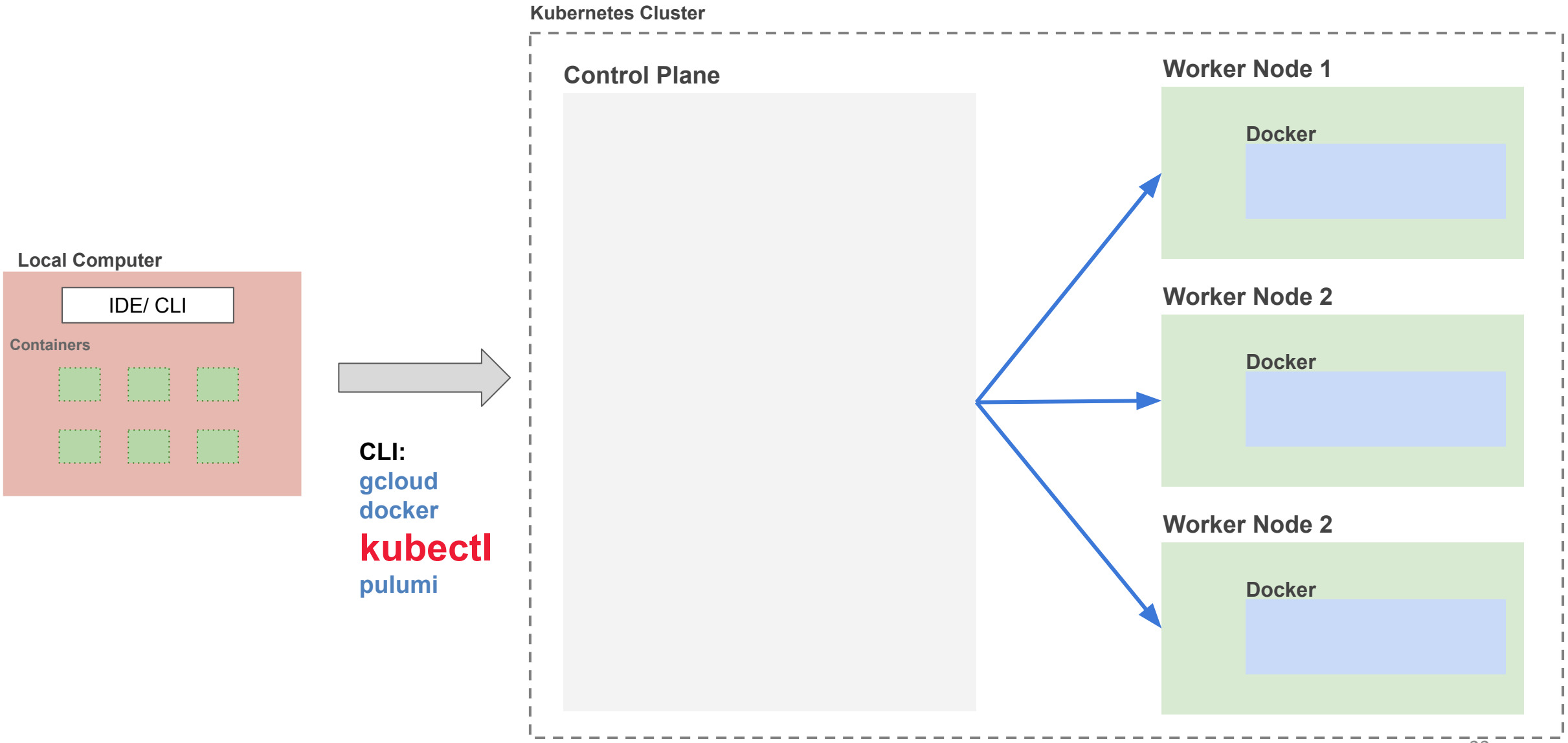
- **Automating** and **Management** of Microservices
- **Bridging** Application Deployment & Deployment (Dev + Ops)
- **Standardizing** Cloud Deployments
- Daily **Management** of Applications

How do we build with Kubernetes?

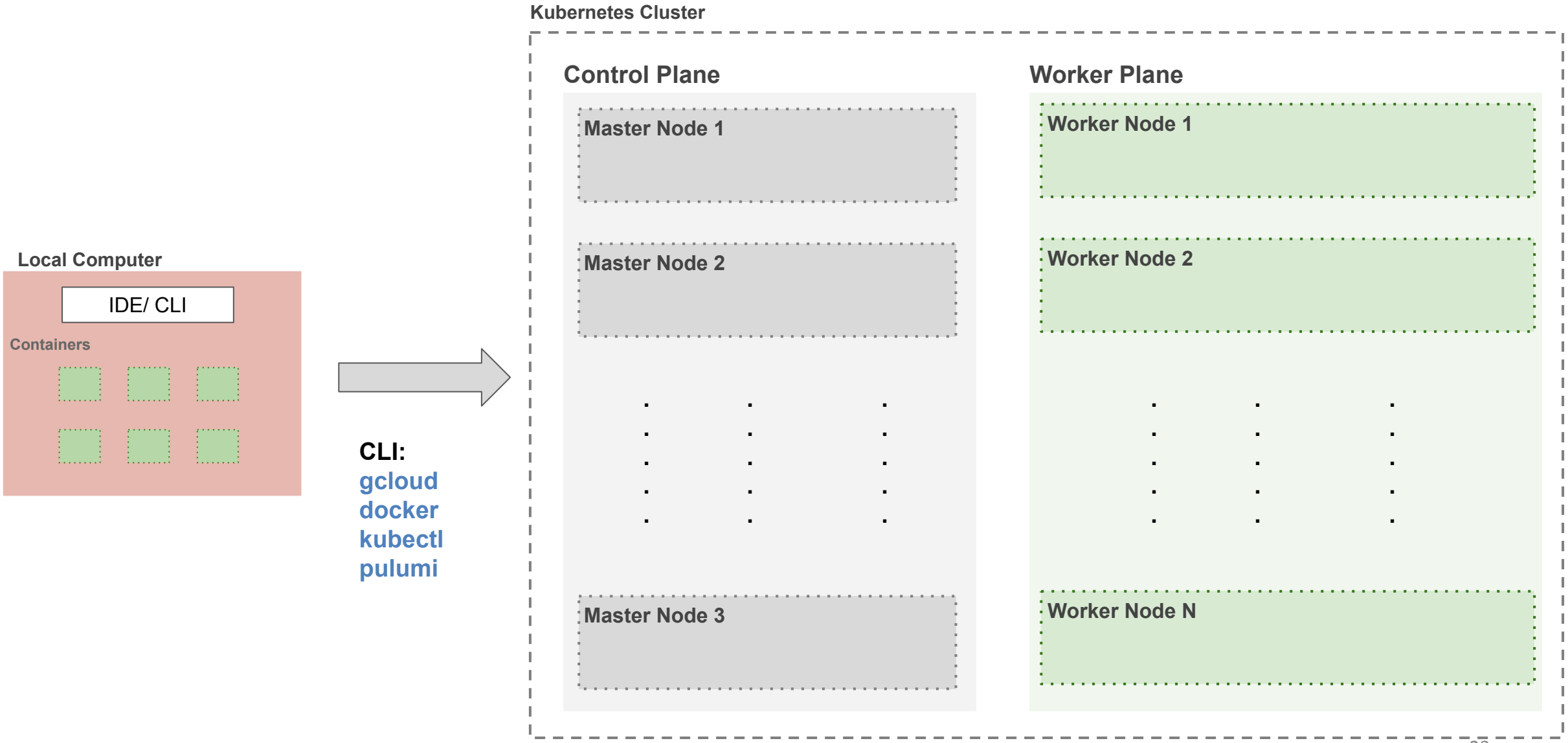
Remember the Cheese App Architecture:



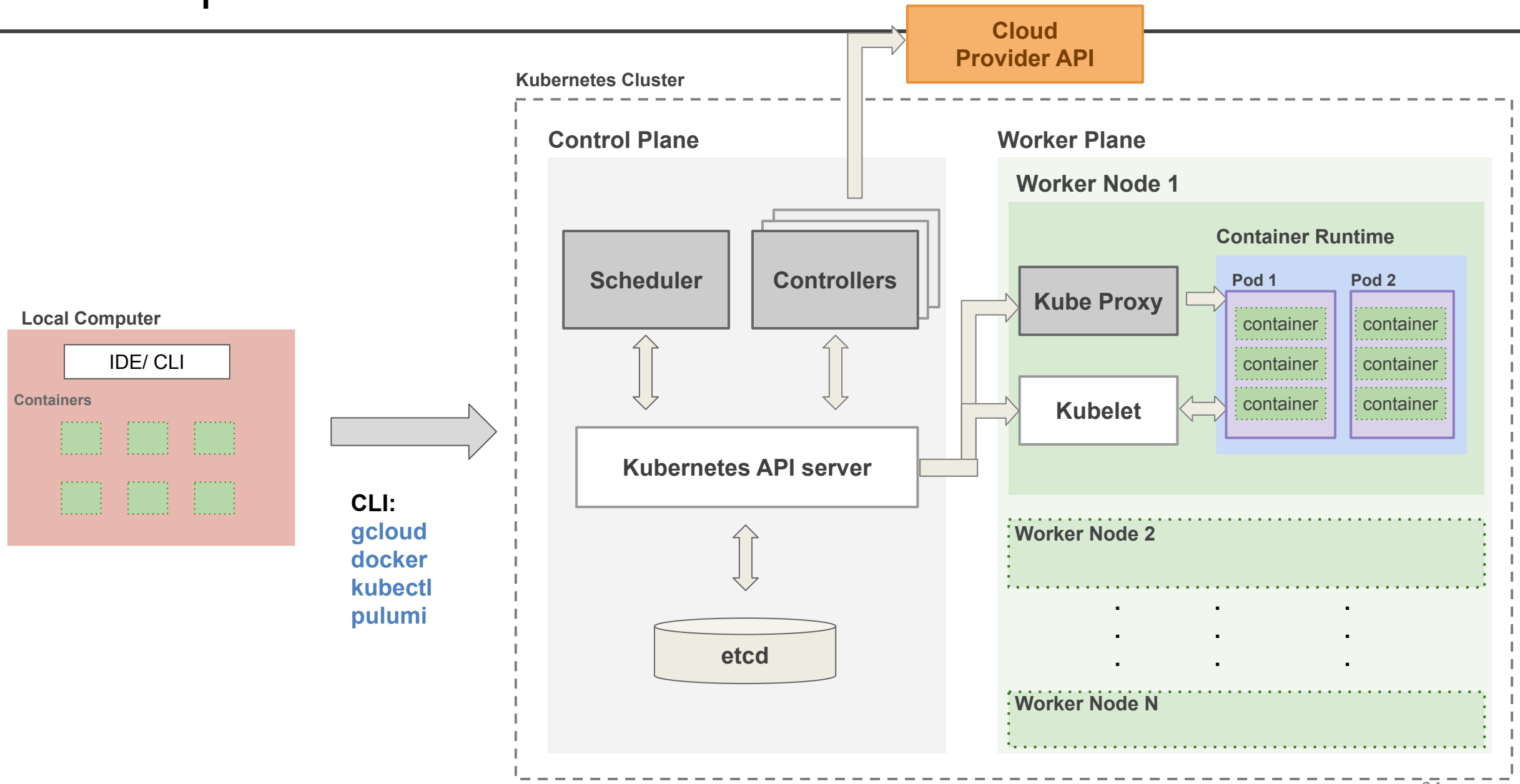
How do we build with Kubernetes?



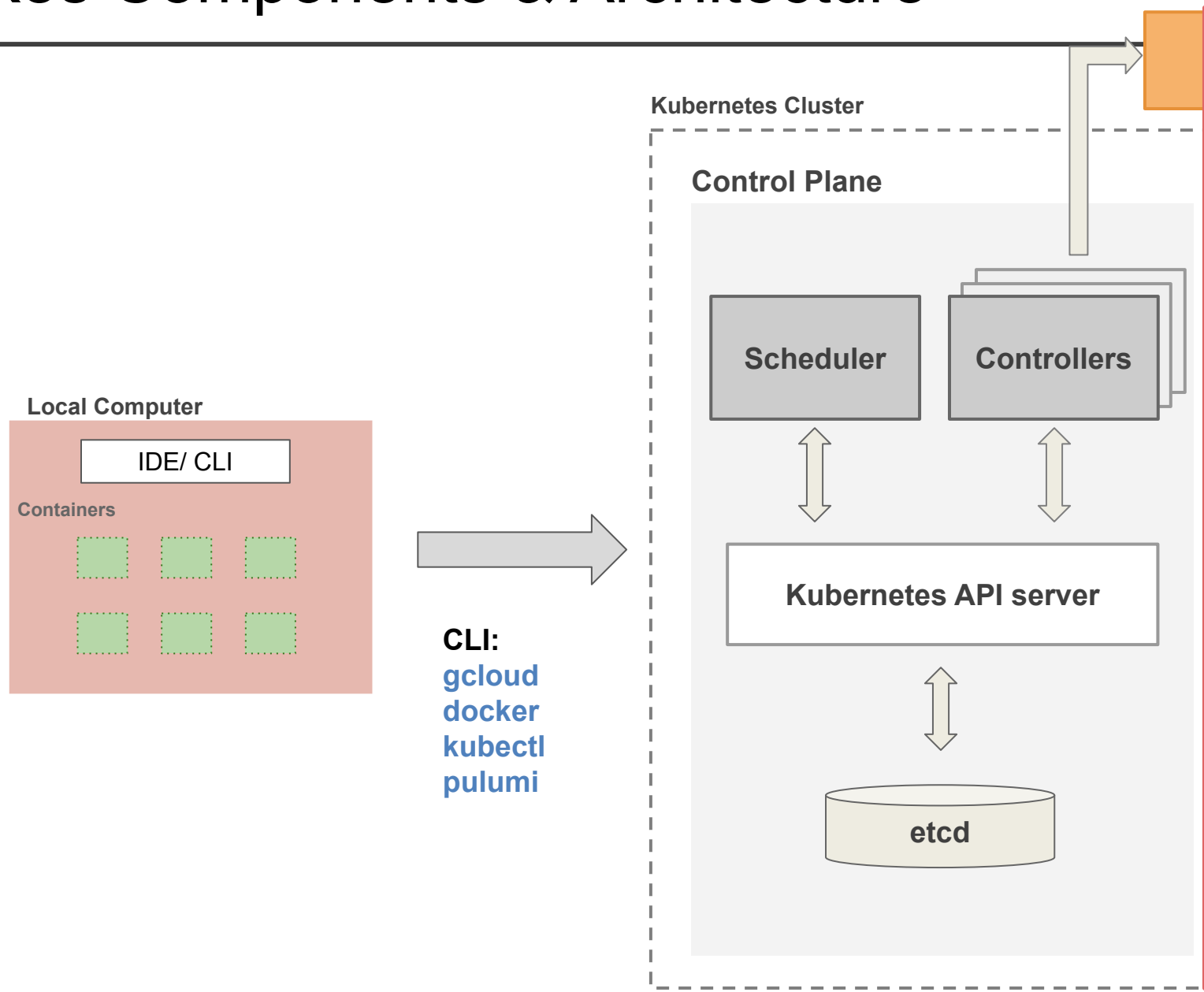
K8s Components & Architecture



K8s Components & Architecture



K8s Components & Architecture



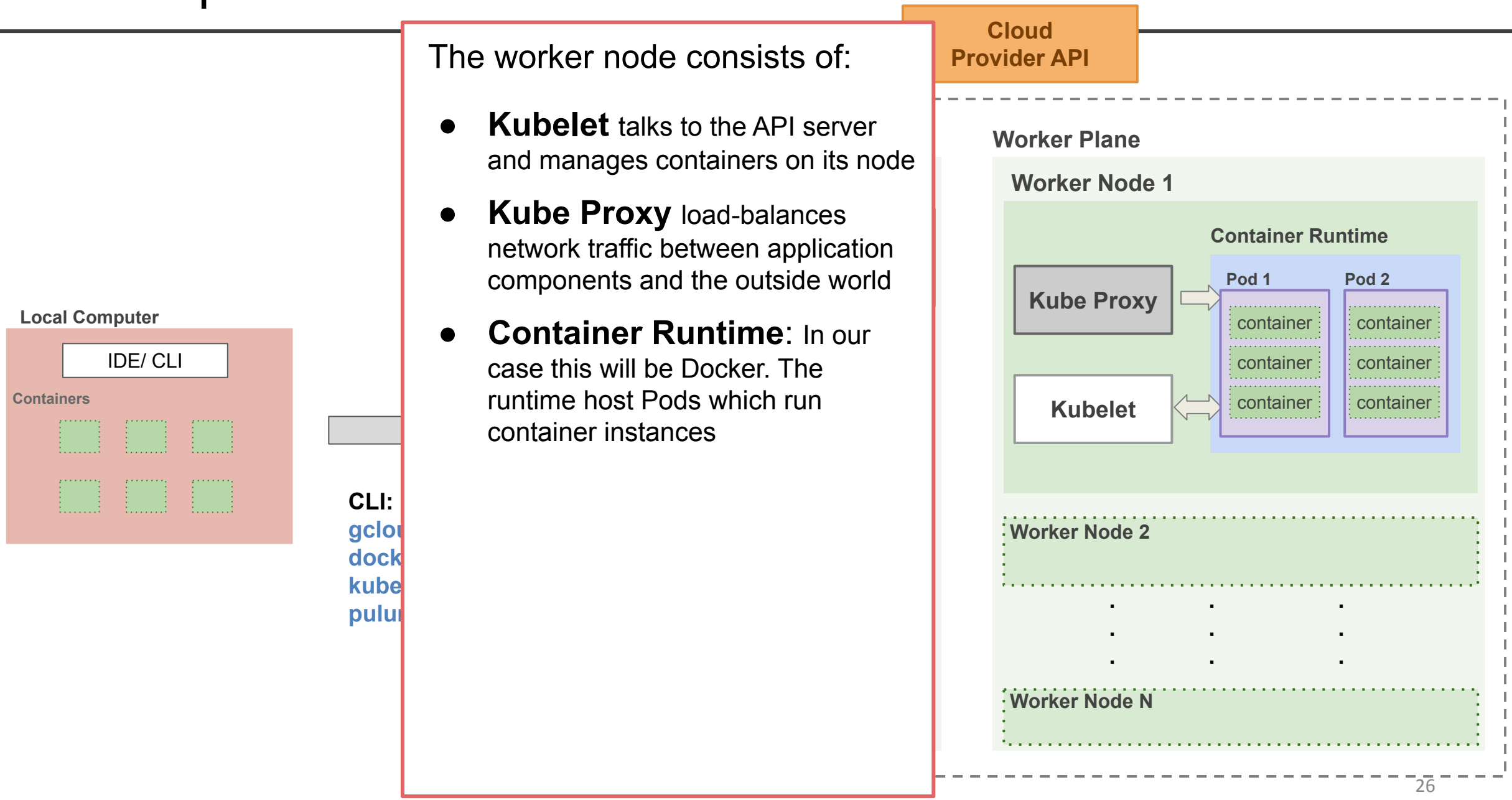
The control plane has:

- **API server** contains various methods to directly access the Kubernetes
- **etcd** works as backend for service discovery that stores the cluster's state and its configuration
- **Scheduler** assigns applications to each worker node
- **Controller manager**:
  - Keeps track of worker nodes
  - Handles node failures and replicates if needed
  - Provide endpoints to access the application from the outside world
  - Communicates with cloud provide regarding resources such as nodes and IP addresses

K8s Components & Architecture

The worker node consists of:

- **Kubelet** talks to the API server and manages containers on its node
- **Kube Proxy** load-balances network traffic between application components and the outside world
- **Container Runtime**: In our case this will be Docker. The runtime host Pods which run container instances



kubectl

Kubernetes provides a command line tool **kubectl** for communicating with a Kubernetes cluster's control plane, using the Kubernetes API.

Use the following syntax to run kubectl commands from your terminal window:

```
kubectl [command] [TYPE] [NAME] [flags]
```

kubectl

```
# Create a service using the definition in example-service.yaml.
kubectl apply -f emoji.yaml

#See what was created
kubectl get deployments
kubectl get pods
kubectl get svc
```

kubectl

Use port-forwarding to map the Kubernetes service to a local port so you can open it in your browser.

```
kubectl port-forward svc/emojirain 8080:80
```

Delete all pods in the default namespace.

```
kubectl delete pods --all
```

Apply all resource definitions found in a given directory.

```
kubectl apply -f <directory>
```

Kubernetes Summary

- **Abstracting** Infrastructure
- **Standardize** Application Deployment
- Deploy Applications **Declaratively**
- Daily **Management** of Applications

Outline

1. Motivation
2. Introduction to Kubernetes
3. **Tutorial: Deploying a Kubernetes Cluster**
4. Advantages of using Kubernetes

Create Kubernetes Cluster

To create a Kubernetes cluster

- You must first install ***gcloud*** which is the GCPs command-line tool
- You create and delete clusters using ***gcloud***

Example:

Create a 1 node Kubernetes Cluster

```
gcloud container clusters create test-cluster --num-nodes 1 --zone us-east1-c
```

Creating cluster test-cluster in us-east1-c...::

Create Kubernetes Cluster

Create a 1 node Kubernetes Cluster

```
gcloud container clusters create test-cluster --num-nodes 1 --zone us-east1-c
```

To inspect the contents of your cluster, go to: https://console.cloud.google.com/kubernetes/...
kubeconfig entry generated for test-cluster.

NAME	LOCATION	MASTER_VERSION	MASTER_IP	MACHINE_TYPE	NODE_VERSION	NUM_NODES	STATUS
test-cluster	us-east1-c	1.20.9-gke.701	34.73.126.138	e2-medium	1.20.9-gke.701	1	RUNNING

Connect to Kubernetes Cluster

```
gcloud container clusters get-credentials test-cluster --zone us-east1-c
--project ac215-project
```

Deploying to Kubernetes Cluster

To create a Kubernetes cluster and deploy app to it.

- You must first install ***kubectl*** which is the Kubernetes command-line tool
- You can manage all resources in Kubernetes using ***kubectl***

Examples:

Get version of client

```
kubectl version --client
```

Client Version: version.Info{Major:"1", Minor:"22", GitVersion:"v1.22.1", GitCommit:"632ed300f2c34f6d6d15ca4cef3d3c7073412212", GitTreeState:"clean", BuildDate:"2021-08-19T15:45:37Z", GoVersion:"go1.16.7", Compiler:"gc", Platform:"linux/amd64"}

Get version of server

```
kubectl version
```

Client Version: version.Info{Major:"1", Minor:"22", GitVersion:"v1.22.1", GitCommit:"632ed300f2c34f6d6d15ca4cef3d3c7073412212", GitTreeState:"clean", BuildDate:"2021-08-19T15:45:37Z", GoVersion:"go1.16.7", Compiler:"gc", Platform:"linux/amd64"}
The connection to the server localhost:8080 was refused - did you specify the right host or port?

Deploying to Kubernetes Cluster

Examples:

Get Kubernetes Cluster Information

```
kubectl get all
```

```
NAME                 TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
service/kubernetes   ClusterIP   10.3.240.1   <none>        443/TCP   48m
```

Get Kubernetes Component Status

```
kubectl get componentstatuses
```

```
NAME                 STATUS    MESSAGE             ERROR
scheduler            Healthy   ok
etcd-1               Healthy   {"health":"true"}
controller-manager   Healthy   ok
etcd-0               Healthy   {"health":"true"}
```

Deploying to Kubernetes Cluster

Examples:

Get Kubernetes Cluster Nodes

```
kubectl get nodes
```

```
NAME                                        STATUS   ROLES    AGE   VERSION
gke-test-cluster-default-pool-2e9eafc9-kj0s   Ready    <none>   51m   v1.20.9-gke.701
```

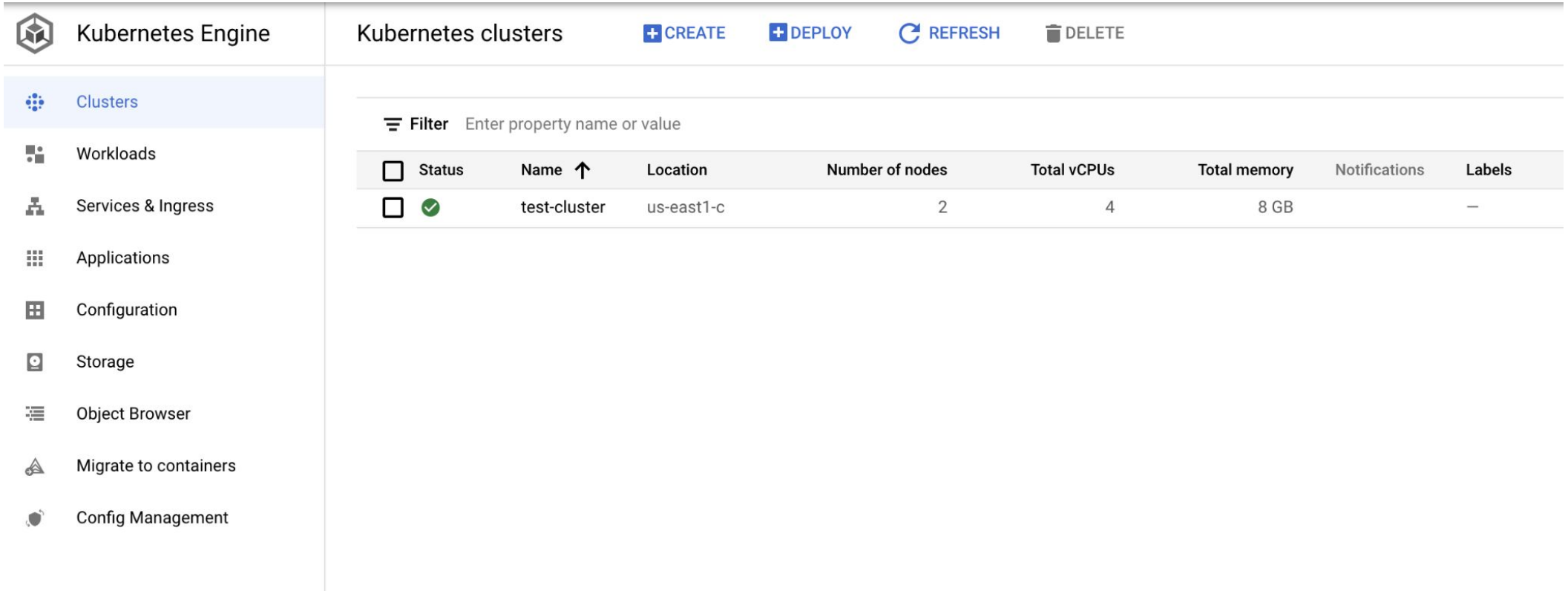
Get Kubernetes Pods

```
kubectl get pods
```

```
NAME                   READY   STATUS    RESTARTS   AGE
web-854b4968b9-j4x8f   1/1     Running   0          49m
web-854b4968b9-x24jh   1/1     Running   0          49m
```

Deploying to Kubernetes Cluster

You can view Kubernetes cluster details directly from GCP



Deploying to Kubernetes Cluster

Examples:

Deploy App to Kubernetes

```
kubectl apply -f deploy-k8s-tic-tac-toe.yml
```

deployment.apps/web created
service/web created

Get Services

```
kubectl get services
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.3.240.1	<none>	443/TCP	29m
web	LoadBalancer	10.3.242.77	**34.139.195.206**	80:32088/TCP	3m51s

Deploying to Kubernetes Cluster

Deployment YAML

```yaml
---
apiVersion: apps/v1
kind: Deployment
spec:
  replicas: 2
  containers:
   - image: dlops/tic-tac-toe
     imagePullPolicy: IfNotPresent
     name: web
     ports:
      - containerPort: 8080
        protocol: TCP
```

Deployment:

- Decares what is in a pod and how many replicas
- Is in charge of keeping the pod running

Service YAML

```yaml
---
apiVersion: v1
kind: Service
spec:
 ports:
 - port: 80
   protocol: TCP
   targetPort: 8080
 type: LoadBalancer
```

Service:

- Decares how traffic is routed to a pod or a multiple replicas.
- Service allows pods to die

Deleting a Kubernetes Cluster

Example:

Delete Kubernetes Cluster called test-cluster

```
gcloud container clusters delete test-cluster --zone us-east1-c
```

The following clusters will be deleted.
 - [test-cluster] in [us-east1-c]

Do you want to continue (Y/n)? Y

Deleting cluster test-cluster...done.
Deleted [https://container.googleapis.com/v1/projects/.../zones/us-east1-c/clusters/test-cluster].

Tutorial: Deploying a Kubernetes Cluster

Deploying a Kubernetes Cluster

Run an pulumi code now for the cheese app.

https://github.com/dlops-io/cheese-app-v3?tab=readme-ov-file#deployment-with-scaling-using-kubernetes



Outline

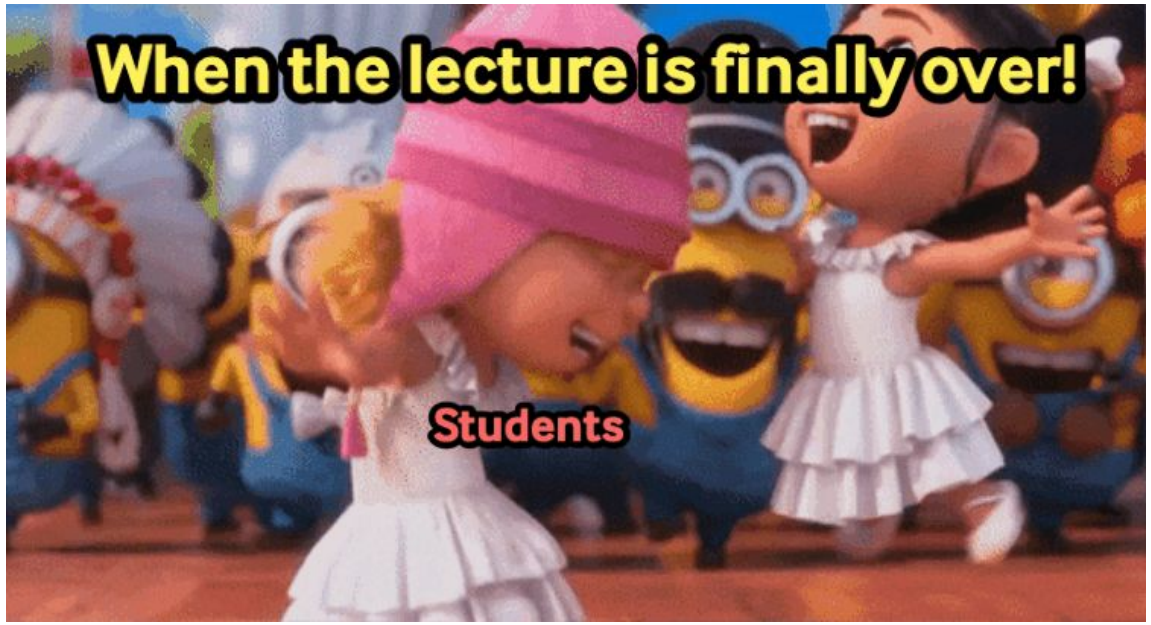
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Advantages of using Kubernetes

- **Self-Service** Deployment of Applications
- **Reduce Cost** by better Infrastructure Utilization
- **Automatically Adjusting** to varying loads
- Running Applications **Smoothly**
- Simplifying Application **Development**



DON’T FORGET TO DELETE


When the lecture is finally over!
Students